

■ Multi-Agent System Architecture (2025 AI Technologies)

Layer	AI Technologies	Role in Multi-Agent System
Core Agents (Specialists)	Generative AI / LLMs Multimodal AI Vision-Language-Action (VLA) Domain-Specific AI	Specialist agents that create, understand, and act.
System Intelligence (Coordinator)	Agentic AI / Autonomous Agents Multi-Agent Systems Embodied AI	Managers that plan, execute, and give the system a
Infrastructure & Governance (Support)	Edge / On-Device AI Long-Context Models AI Infrastructure Explainable & Trustworthy AI Sustainable / Green AI Hyper-Personalization & Ambient AI AI-Security Integration	The foundation: memory, compute, safety, efficiency

■ Example: Autonomous Car as a Multi-Agent System

- **Core agents:** Vision AI (cameras), Language AI (navigation), Control AI (steering/braking)
- **System intelligence:** Agentic planning (routes), Multi-agent coordination (cars talking)
- **Infrastructure:** Edge AI (chips in car), Long-context (trip memory), Explainable AI (safety reports)

■ Multi-Agent AI System Architecture (Layered Diagram)

Core Agents (Specialists):

- Generative AI / LLMs
- Multimodal AI
- Vision-Language-Action (VLA)
- Domain-Specific AI

System Intelligence (Coordinators):

- Agentic AI / Autonomous Agents
- Multi-Agent Systems
- Embodied AI

Infrastructure & Governance (Support):

- Edge / On-Device AI
- Long-Context Models
- AI Infrastructure
- Explainable & Trustworthy AI
- Sustainable / Green AI
- Hyper-Personalization & Ambient AI
- AI-Robotics Integration

Example: Autonomous Car = Core (vision, control) + System Intelligence (planning) + Infrastructure (on-device chips).

■ Multi-Agent AI System Notes

This document summarizes the 14 key AI technologies of 2025 and shows how they fit together inside a Multi-Agent System architecture. The technologies are divided into three layers: Core Agents, System Intelligence, and Infrastructure & Governance.

■ Core Agents (Specialists)

- Generative AI / LLMs → Create text, images, and code (e.g., ChatGPT, MidJourney).
- Multimodal AI → Understands text, images, audio, and video together (e.g., GPT-4o, Gemini).
- Vision-Language-Action (VLA) → Sees, understands, and acts (e.g., robotics models like Helix, GR00T).
- Domain-Specific AI → Specialized expert AIs for medicine, law, finance (e.g., Med-PaLM, BloombergGPT).

■ System Intelligence (Coordinators)

- Agentic AI → Plans and works on tasks independently (e.g., AutoGPT, Copilot agents).
- Multi-Agent Systems → Multiple AIs collaborate like a team (e.g., Stanford Smallville, AutoGen).
- Embodied AI → AI in a physical body like a robot, car, or drone (e.g., Tesla Optimus, Boston Dynamics).

■ Infrastructure & Governance (Support)

- Edge / On-Device AI → Runs locally on phones/robots without internet (e.g., Apple Intelligence, Pixel Nano).
- Long-Context Models → Handle very long memory (e.g., Claude 3.5, Gemini 1.5).
- AI Infrastructure → GPUs, TPUs, and supercomputers powering AI (e.g., NVIDIA Blackwell, Google TPUs).
- Explainable & Trustworthy AI → Shows reasons for AI decisions (e.g., IBM OpenScale, EU AI Act).
- Sustainable / Green AI → Energy-efficient AI (e.g., Google carbon-neutral AI).
- Hyper-Personalization & Ambient AI → AI adapts to your life, runs in background (e.g., Spotify AI, Apple).
- AI-Robotics Integration → Smarter robots powered by AI (e.g., Amazon robots, Agility Digit).

■ Example: Autonomous Cars

Autonomous cars are a combination of many AI technologies:

- Core: Embodied AI + VLA + Edge AI (car sees, plans, and acts in real time).
- System Intelligence: Agentic AI for route planning, Multi-Agent Systems for car-to-car coordination.
- Infrastructure: Long-context memory, Explainable AI (safety), and domain-specific driving AI.